

Problem

The voltage across a 1.1-kW toaster that produces a current of 10 A is:

- (a) 11kV
- (b) 1100 V
- (c) 110V
- (d) 11V

Step-by-step solution

Step 1 of 2

The power absorbed or supplied by an element is the product of the voltage across the element and the current through it.

$$p = vi$$

Step 2 of 2

Substitute 1.1×10^3 for p and 10 for i in the equation.

$$1.1 \times 10^3 = v(10)$$

Solve for voltage v in the equation.

$$\begin{aligned} v &= \frac{1.1 \times 10^3}{10} \\ &= 110 \text{ V} \end{aligned}$$

Thus, voltage across toaster v is 110 V

So, correct option is ☒ c.